SYSTEM REQUIREMENTS SPECIFICATION
STUDENT MANAGEMENT SYSTEM

Version 1.0

Document Change Log

Ngày	Phiên bản	Mô tả	Tác giả
20/10/26	1.0	Initial document	
30/10/26	1.1	Update Use-case Login	
31/10/26	1.2	Update requirement	

I. Student Management System

The university needs a student management system with the following basic features:

1. Account Management

- Students, Teachers, and the Administrator log in and out of the system. Students log in with their student ID and password. Teachers log in with teacher ID or email and password.
- Students can view their information and update only their contact details (email, phone number, address). Teachers can update their phone number and email. Other information can only be changed by the Administrator.
- If a user forgets the password, they can request a reset via email or ask the Administrator.
- The Administrator can create, edit, lock, and unlock Student and Teacher accounts.
- Access is restricted by role: each user can only use the functions and see the data they are permitted to.

2. Student Information Management

- The Administrator adds student records: student ID, full name, date of birth, gender, email, phone number, address, place of birth, academic cohort, major, and student class.

- The Administrator can edit student information and academic status (Enrolled, Withdrawn, Temporarily Suspended, Graduated).
- The Administrator can view all students. Teachers can view students in the course sections assigned to them.
- The system shows clear messages when an operation fails.

3. Student Search And Filtering

- The Administrator and Teachers can search students by student ID, full name (partial or complete), student class, or major. Teachers only see students in their assigned course sections.
- The Administrator and Teachers can filter by student class, academic cohort, major, and academic status.
- Results show student ID, full name, student class, major, and academic status. If nothing matches, the system shows a notification.
- The Administrator and Teachers can also open a student's detailed record from the results.

4. Course and course section management

- The Administrator manages courses: course code, course name, credit hours, number of class periods. Course codes must be unique.
- The Administrator creates course sections for each semester: course, class schedule, maximum capacity, and assigned Teacher.
- The Administrator can view the student list of any course section. Teachers can view the lists of their assigned sections.
- Students, Teachers, and the Administrator can view the course sections offered each semester

5. Course registration management

- Students view the course sections open for registration, including course name, code, credit hours, Teacher, schedule, and remaining capacity.
- Students register for a course section during the registration period. The system checks the registration period, remaining capacity, duplicate registration, and prerequisites. Invalid requests are rejected with a reason.
- Students can view their registered sections and cancel a registration before the cancellation deadline.

- The system updates remaining capacity after each registration or cancellation and ensures the number of students never exceeds the maximum capacity

6. Grade and academic result management

- Assigned Teachers enter and edit grades for Students in their course sections while the gradebook is not locked.
- Grade components: attendance, midterm, and final examination, each with a weight (total 100%). The system automatically calculates the final course grade.
- The system converts the final grade into a letter grade (A, B, C, D, F) and a four-point scale, and calculates the semester GPA (average of four-point grades weighted by credit hours).
- The system checks that grades are within the valid range before saving.
- The Teacher can lock the gradebook after grading is finished. Only the Administrator can unlock it.
- Students can view their own component grades, final grades, and semester GPA. Other users cannot view or change them without permission.

II. Supplementary Specification

1. Usability

- The system shall provide a simple and user-friendly interface.
- The system shall provide clear instructions for users to perform each function.
- The system shall display clear error messages when users enter invalid information.
- The system shall allow users to navigate between functions easily.
- The system shall provide an appropriate interface for each user role: Administrator, Teacher, and Student.

2. Reliability

- The system shall store student information, course information, and academic results accurately.
- The system shall validate input data before saving it to the database.
- The system shall prevent duplicate student IDs and course codes.
- The system shall display an error message when an operation fails.
- The system shall not modify existing data if an update operation fails.

3. Performance

- The system shall respond to common operations, such as login, student search, and viewing academic results, within 3 seconds under normal operating conditions.
- The system shall support 500 users accessing the system simultaneously.

4. Security

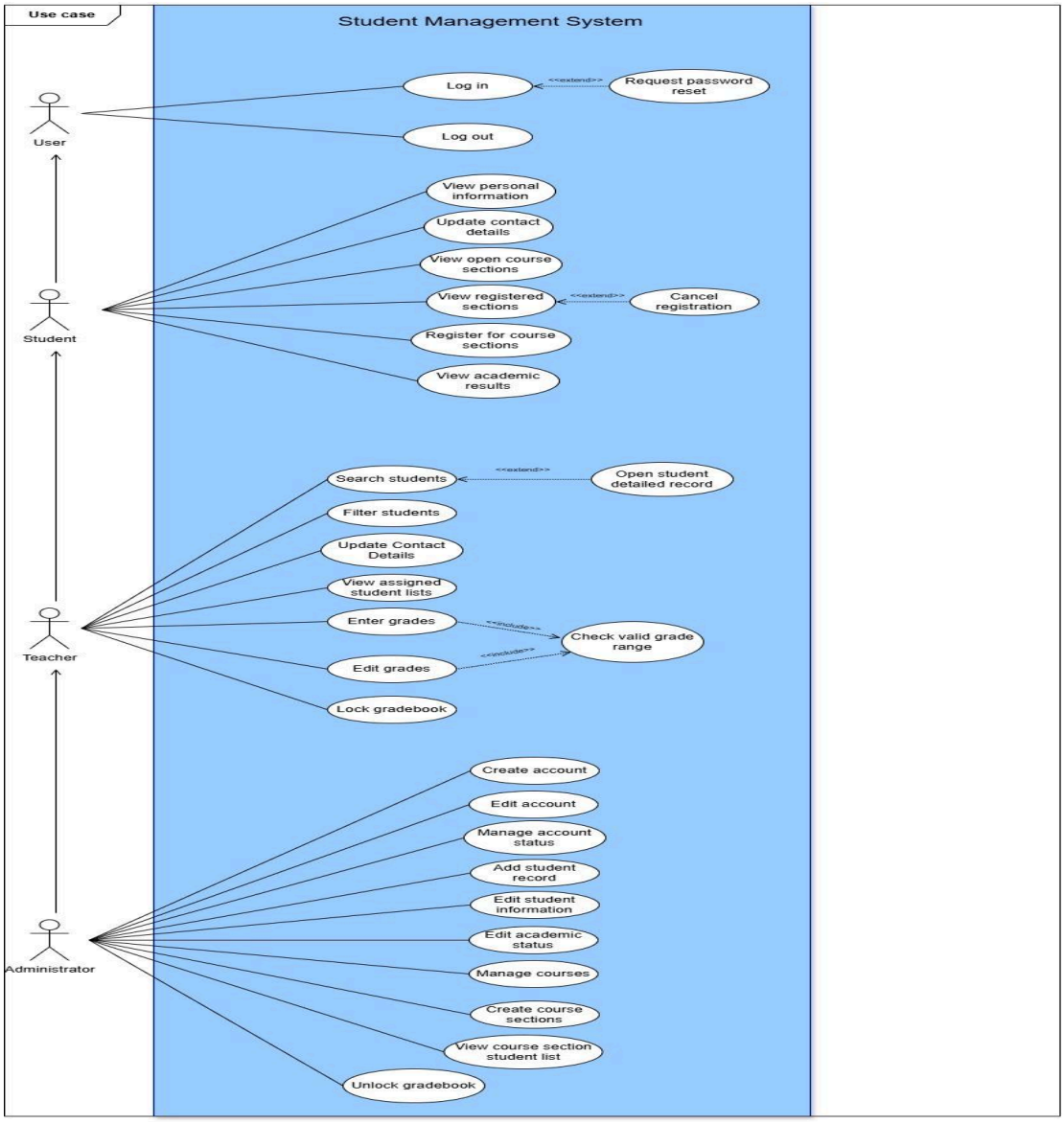
- The system shall assign access permissions according to user roles.
- The Administrator shall be allowed to manage student information, course information, and user accounts.
- The Teacher shall be allowed to manage grades only within the permissions granted to them.
- The Student shall only be allowed to view their own academic results.

5. Development

- The system shall be developed using the Client–Server architecture, in which clients send requests to the server and the server processes those requests.
- The client shall communicate with the server to perform functions such as logging in, searching for students, managing courses, and viewing academic results.
- The server shall process business logic and communicate with the database to retrieve or update data..
- The server shall validate input data and verify user permissions before performing operations involving system data.

III. Use-Case Diagram

1. Overview



1.1. List of Actors

No.	Actor	Description
1	User	Generalized actor representing any person who uses the system (Student, Teacher or Administrator); logs in, logs out and can request a password reset
2	Student	A student who has an account and is logged in; views personal information, registers for course sections and views academic results
3	Teacher	A teacher who views assigned student lists, enters and edits grades, and locks the gradebook
4	Administrator	The system administrator who manages accounts, student records, courses and course sections, and unlocks gradebooks

1.2. List of Use-cases

No.	Use-case	Description
1	Log in	Log in to the system
2	Log out	Log out of the system
3	Request password reset	Request a password reset
4	View personal information	View personal information
5	Update contact details	Update contact details
6	View open course sections	View open course sections
7	Register for course sections	Register for course sections

8	View registered sections	View registered course sections
9	Cancel registration	Cancel a course section registration
10	View academic results	View academic results
11	View assigned student lists	View the lists of assigned students
12	Filter students	Filter students
13	Open student detailed record	Open a student's detailed record
14	Enter grades	Enter grades
15	Edit grades	Edit grades
16	Check valid grade range	Check that grades are within the valid range
17	Lock gradebook	Lock the gradebook
18	Unlock gradebook	Unlock the gradebook
19	Create account	Create an account
20	Edit account	Edit an account
21	Manage account status	Manage account status
22	Search students	Search for students
23	Add student record	Add a student record
24	Edit student information	Edit student information
25	Edit academic status	Edit academic status
26	Manage courses	Manage courses

27	Create course sections	Create course sections
28	View course section student list	View the student list of a course section

2. Use-case Log in

2.1. Summary

This use-case describes how a user (Student, Teacher or Administrator) logs in to the Student Management System.

2.2. Flow of events

Main flow

The use-case starts when a user wants to log in to the system.

1. The system asks the user to enter a username and password.
2. The user enters the username and password and requests to log in.
3. The system checks the entered information. If it is correct, the system lets the user log in and displays the functions that match the user's role (Student, Teacher or Administrator).

Alternative flows

- Wrong username or password: the system displays an error message. The user can go back to the start of the main flow or cancel the login, at which point the use-case ends.
- Account locked or inactive: the system reports that the account cannot be used and the use-case ends.

2.3. Special requirements

Passwords must be stored in encrypted form.

2.4. System state before the use-case starts

The user is not logged in.

2.5. System state after the use-case ends

If the use-case succeeds, the user is logged in. Otherwise, the system state does not change.

2.6. Extension points

Use-case Request password reset: if the user has forgotten the password.

3. Use-case Log out

3.1. Summary

This use-case allows a logged-in user to leave the system.

3.2. Flow of events

Main flow

The use-case starts when the user selects the log out function.

1. The system ends the user's session.
2. The system redirects the user to the login page.

Alternative flows

None.

3.3. Special requirements

None.

3.4. System state before the use-case starts

The user must be logged in.

3.5. System state after the use-case ends

The user is not logged in.

3.6. Extension points

None.

4. Use-case Request password reset

4.1. Summary

This use-case allows a user (Student, Teacher or Administrator) to request a password reset when the password has been forgotten. It extends the use-case Log in.

4.2. Flow of events

Main flow

The use-case starts when the user selects the password reset function while logging in.

1. The system asks the user to enter the registered email or username (student ID or teacher ID).
2. The user enters the information and submits the request.
3. The system checks whether the information exists in the system. If it does, the system sends instructions or a new password to the user's email.
4. The system reports the result of the request.

Alternative flows

The information does not exist in the system: the system displays an error message. The user can re-enter the information or exit the function.

4.3. Special requirements

The new password must be randomly generated and stored in encrypted form.

4.4. System state before the use-case starts

The user is not logged in.

4.5. System state after the use-case ends

If the use-case succeeds, the user's password is reset. Otherwise, the system state does not change.

4.6. Extension points

None.

5. Use-case View personal information

5.1. Summary

This use-case allows a student to view their own personal information.

5.2. Flow of events

Main flow

The use-case starts when the student selects the function to view personal information.

1. The system displays the student's personal information: student ID, full name, date of birth, gender, address, phone number and email.

Alternative flows

None.

5.3. Special requirements

A student can only view their own information.

5.4. System state before the use-case starts

The student must be logged in.

5.5. System state after the use-case ends

None.

5.6. Extension points

None.

6. Use-case Update contact details

6.1. Summary

This use-case allows a student or a teacher to update their contact details (student: address, phone number, email; teacher: phone number, email).

6.2. Flow of events

Main flow

The use-case starts when the user selects the function to update contact details.

1. The system displays the user's current contact details.
2. The user edits the contact details as desired.
3. The user requests the update.
4. The system validates the information and updates it in the system.
5. The system reports the result of the update.

Alternative flows

- The user cancels: the system does not update the information and the use-case ends.
- Invalid information or missing required fields: the system displays an error message. The user can correct the information or cancel the update.

6.3. Special requirements

A student can update address, phone number and email; a teacher can only update phone number and email. Neither can change academic information.

6.4. System state before the use-case starts

The user (Student or Teacher) must be logged in.

6.5. System state after the use-case ends

If the use-case succeeds, the contact details are updated. Otherwise, the system state does not change.

6.6. Extension points

None.

7. Use-case View open course sections

7.1. Summary

This use-case allows a student to view the list of course sections that are open for registration.

7.2. Flow of events

Main flow

The use-case starts when the student selects the function to view open course sections.

1. The system displays the list of open course sections: section code, course name, teacher, schedule, maximum capacity and remaining seats.

Alternative flows

No course section is open: the system displays a corresponding message.

7.3. Special requirements

The list of course sections must be paginated when it is large.

7.4. System state before the use-case starts

The student must be logged in.

7.5. System state after the use-case ends

None.

7.6. Extension points

None.

8. Use-case Register for course sections

8.1. Summary

This use-case allows a student to register for one or more open course sections.

8.2. Flow of events

Main flow

The use-case starts when the student selects the function to register for course sections.

1. The student selects the open course section(s) to register for.

2. The system checks the registration conditions: the section still has seats, the student has not already registered for it, and its schedule does not conflict with the sections already registered.
3. The system asks the student to confirm the registration.
4. If the student confirms, the system records the registration and reduces the remaining seats of the section.
5. The system reports the result of the registration.

Alternative flows

- The section is full: the system reports this and does not allow registration.
- Schedule conflict or prerequisites not met: the system displays an error message. The student can choose another section or stop registering.
- The student does not confirm: the system does not record the registration.

8.3. Special requirements

Registration must be handled correctly when many students register at the same time for a section with few seats left, so that the maximum capacity is never exceeded.

8.4. System state before the use-case starts

The student must be logged in.

8.5. System state after the use-case ends

If the use-case succeeds, the section is added to the student's list of registered sections. Otherwise, the system state does not change.

8.6. Extension points

None.

9. Use-case View registered sections

9.1. Summary

This use-case allows a student to view the list of course sections they have registered for.

9.2. Flow of events

Main flow

The use-case starts when the student selects the function to view registered sections.

1. The system displays the student's registered sections: section code, course name, teacher, schedule and registration date.
2. If the student chooses to cancel a registration, the use-case Cancel registration is performed.

Alternative flows

The student has not registered for any section: the system displays a corresponding message.

9.3. Special requirements

A student can only view their own sections.

9.4. System state before the use-case starts

The student must be logged in.

9.5. System state after the use-case ends

None.

9.6. Extension points

Use-case Cancel registration: if the student chooses to cancel a registration.

10. Use-case Cancel registration

10.1. Summary

This use-case allows a student to cancel the registration of a course section they have registered for.

10.2. Flow of events

Main flow

The use-case starts when the student chooses to cancel the registration of a section in the list of registered sections.

1. The system displays a dialog asking the student to confirm the cancellation.
2. If the student agrees, the system checks that the cancellation is valid (the section belongs to the student and the cancellation deadline has not passed).
3. The system cancels the registration and increases the remaining seats of the section.
4. The system reports the result of the cancellation.

Alternative flows

- The student does not agree: the system changes nothing and the use-case ends.
- The cancellation deadline has passed: the system displays an error message and does not allow the cancellation.

10.3. Special requirements

A student can only cancel their own sections.

10.4. System state before the use-case starts

The student must be logged in.

10.5. System state after the use-case ends

If the use-case succeeds, the section is removed from the student's registrations. Otherwise, the system state does not change.

10.6. Extension points

None.

11. Use-case View academic results

11.1. Summary

This use-case allows a student to view their own academic results.

11.2. Flow of events

Main flow

The use-case starts when the student selects the function to view academic results.

1. The system displays the results for each course section: section code, course name, credits, grade, and pass or fail result.
2. The system displays the student's grade point average and current academic status.

Alternative flows

There are no results yet: the system displays a corresponding message.

11.3. Special requirements

A student can only view their own results. Only grades that the teacher has locked are displayed.

11.4. System state before the use-case starts

The student must be logged in.

11.5. System state after the use-case ends

None.

11.6. Extension points

None.

12. Use-case View assigned student lists

12.1. Summary

This use-case allows a teacher to view the lists of students in the course sections they are assigned to teach.

12.2. Flow of events

Main flow

The use-case starts when the teacher selects the function to view assigned student lists.

1. The system displays the list of course sections the teacher is assigned to.
2. The teacher selects a course section.
3. The system displays the list of students in that section: student ID, full name and class.

Alternative flows

The teacher has not been assigned any course section: the system displays a corresponding message.

12.3. Special requirements

A teacher can only view the student lists of the sections they are assigned to. The list must be paginated.

12.4. System state before the use-case starts

The teacher must be logged in.

12.5. System state after the use-case ends

None.

12.6. Extension points

None.

13. Use-case Filter students

13.1. Summary

This use-case allows a teacher or an administrator to filter students by certain criteria. A teacher only sees students in the course sections assigned to them.

13.2. Flow of events

Main flow

The use-case starts when the user chooses the filter function.

1. The system displays the filter criteria (for example: class, academic cohort, major, academic status).
2. The user selects or enters the criteria and requests the filter.
3. The system displays the list of students that match the selected criteria.

Alternative flows

No student matches the criteria: the system displays a no-results message.

13.3. Special requirements

A teacher can only filter among students in the course sections they are assigned to.

13.4. System state before the use-case starts

The user (Teacher or Administrator) must be logged in.

13.5. System state after the use-case ends

None.

13.6. Extension points

None.

14. Use-case Open student detailed record

14.1. Summary

This use-case allows a teacher or an administrator to open the detailed record of a student selected from the search results. It extends the use-case Search students.

14.2. Flow of events

Main flow

The use-case starts when the user chooses to view the detailed record of a student in the search results.

1. If the user is a teacher, the system checks whether the student belongs to the teacher's assigned course sections.
2. The system displays the student's detailed record: personal information, contact details, current course sections, grades of the sections taught by the teacher, and academic status.

Alternative flows

The student is not in the teacher's assigned course sections: the system reports that the teacher has no permission to view the record and the use-case ends.

14.3. Special requirements

A teacher can only view the records of students in the sections they are responsible for, and cannot edit the record.

14.4. System state before the use-case starts

The user (Teacher or Administrator) must be logged in and the search results of the use-case Search students are displayed.

14.5. System state after the use-case ends

None.

14.6. Extension points

None.

15. Use-case Enter grades

15.1. Summary

This use-case allows a teacher to enter grades for the students of the course sections they are responsible for.

15.2. Flow of events

Main flow

The use-case starts when the teacher selects the function to enter grades for a course section.

1. The system displays the student list of the section and the grade columns to fill in.
2. The teacher enters grades for the students and requests to save.
3. The system performs the use-case Check valid grade range to check each entered grade.
4. If all grades are valid, the system saves the grades.
5. The system reports the result of the grade entry.

Alternative flows

- Some grades are outside the valid range: the system displays an error message and points out the invalid grades. The teacher can correct them or cancel the grade entry.
- The gradebook is locked: the system reports that grades cannot be entered and the use-case ends.

15.3. Special requirements

A teacher can only enter grades for the sections they are responsible for.

15.4. System state before the use-case starts

The teacher must be logged in and the gradebook of the section must not be locked.

15.5. System state after the use-case ends

If the use-case succeeds, the students' grades are saved in the system. Otherwise, the system state does not change.

15.6. Extension points

None. This use-case includes the use-case Check valid grade range.

16. Use-case Edit grades

16.1. Summary

This use-case allows a teacher to edit grades already entered for students, as long as the gradebook is not locked.

16.2. Flow of events

Main flow

The use-case starts when the teacher selects the function to edit grades of a course section.

1. The system displays the list of students with their current grades.
2. The teacher edits the grades of one or more students and requests to save.
3. The system performs the use-case Check valid grade range to check the edited grades.

4. If they are valid, the system updates the grades and records the edit history (editor, time, old grade, new grade).
5. The system reports the result of the update.

Alternative flows

- A grade is outside the valid range: the system displays an error message. The teacher can correct it or cancel the update.
- The gradebook is locked: the system reports that grades cannot be edited. The teacher must ask an administrator to unlock the gradebook.

16.3. Special requirements

Every grade change must be logged for auditing.

16.4. System state before the use-case starts

The teacher must be logged in and the gradebook of the section must not be locked.

16.5. System state after the use-case ends

If the use-case succeeds, the grades are updated and the edit history is saved. Otherwise, the system state does not change.

16.6. Extension points

None. This use-case includes the use-case Check valid grade range.

17. Use-case Check valid grade range

17.1. Summary

This use-case checks whether entered or edited grades are within the valid range of values. It is included by Enter grades and Edit grades.

17.2. Flow of events

Main flow

The use-case starts when the use-case Enter grades or Edit grades requests a grade check.

1. The system checks that each grade is a number within the valid range of the section's grading scale (for example from 0 to 10).

2. The system returns the check result to the calling use-case.

Alternative flows

Some grades are invalid (not a number or outside the allowed range): the system returns the list of invalid grades with an error message.

17.3. Special requirements

The valid range is taken from the configured grading scale of the section.

17.4. System state before the use-case starts

There are grades to check from the use-case Enter grades or Edit grades.

17.5. System state after the use-case ends

The system state does not change.

17.6. Extension points

None.

18. Use-case Lock gradebook

18.1. Summary

This use-case allows a teacher to lock the gradebook of a course section after finishing grade entry.

18.2. Flow of events

Main flow

The use-case starts when the teacher selects the function to lock the gradebook of a course section.

1. The system displays a warning that after locking, the teacher can no longer enter or edit grades, and asks for confirmation.
2. If the teacher confirms, the system checks that grades have been entered for all students of the section.
3. The system changes the gradebook to the "Locked" status and publishes the grades to the students.
4. The system reports the result of locking the gradebook.

Alternative flows

- The teacher does not confirm: the system changes nothing.
- Some students still have no grade: the system reports this and asks the teacher to enter all grades before locking.

18.3. Special requirements

A teacher can only lock the gradebook of the sections they are responsible for.

18.4. System state before the use-case starts

The teacher must be logged in and the gradebook must not be locked.

18.5. System state after the use-case ends

If the use-case succeeds, the gradebook is in the "Locked" status. Otherwise, the system state does not change.

18.6. Extension points

None.

19. Use-case Unlock gradebook

19.1. Summary

This use-case allows an administrator to unlock a locked gradebook so that the teacher can edit grades when needed.

19.2. Flow of events

Main flow

The use-case starts when the administrator selects the function to unlock a gradebook.

1. The system displays the list of locked gradebooks.
2. The administrator selects a gradebook and enters the reason for unlocking.
3. The system asks the administrator to confirm.
4. If the administrator confirms, the system changes the gradebook to the "Unlocked" status and records the history (who unlocked it, when, and why).

5. The system reports the result.

Alternative flows

- The administrator does not confirm: the system changes nothing.
- No reason has been entered: the system asks for a reason before unlocking.

19.3. Special requirements

Every unlock must be recorded in the history.

19.4. System state before the use-case starts

The administrator must be logged in.

19.5. System state after the use-case ends

If the use-case succeeds, the gradebook is unlocked and the teacher can edit grades. Otherwise, the system state does not change.

19.6. Extension points

None.

20. Use-case Create account

20.1. Summary

This use-case allows an administrator to create a login account for a student, a teacher or an administrator.

20.2. Flow of events

Main flow

The use-case starts when the administrator selects the function to create an account.

1. The system asks the administrator to enter the account information: full name, email, role (Student, Teacher, Administrator) and username.
2. The administrator enters the information and requests to create the account.

3. The system validates the information and creates the account with a randomly generated initial password.
4. The system reports the result of the account creation.

Alternative flows

- The username or email already exists: the system displays an error message. The administrator can correct it or cancel.
- Required information is missing: the system displays an error message and asks for the missing fields.

20.3. Special requirements

Passwords must be stored in encrypted form.

20.4. System state before the use-case starts

The administrator must be logged in.

20.5. System state after the use-case ends

If the use-case succeeds, the new account is added to the system. Otherwise, the system state does not change.

20.6. Extension points

None.

21. Use-case Edit account

21.1. Summary

This use-case allows an administrator to edit the information of an existing account.

21.2. Flow of events

Main flow

The use-case starts when the administrator selects the function to edit an account.

1. The system displays the information of the account to be edited.
2. The administrator edits the information (full name, email, role, password reset) as desired.

3. The administrator requests the update.
4. The system validates the information and updates it in the system.
5. The system reports the result of the update.

Alternative flows

- The administrator cancels: the system does not update the information.
- Invalid information or an email already used by another account: the system displays an error message. The administrator can correct it or cancel.

21.3. Special requirements

None.

21.4. System state before the use-case starts

The administrator must be logged in.

21.5. System state after the use-case ends

If the use-case succeeds, the account information is updated. Otherwise, the system state does not change.

21.6. Extension points

None.

22. Use-case Manage account status

22.1. Summary

This use-case allows an administrator to change the status of an account (activate, lock, deactivate).

22.2. Flow of events

Main flow

The use-case starts when the administrator selects the function to manage account status.

1. The system displays the list of accounts with their current status.
2. The administrator selects an account and chooses a new status.

3. The system asks the administrator to confirm.
4. If the administrator confirms, the system updates the account status.
5. The system reports the result.

Alternative flows

- The administrator does not confirm: the system changes nothing.
- The administrator tries to lock their own account: the system reports that this is not allowed.

22.3. Special requirements

A locked or deactivated account cannot log in to the system. The account list must be paginated.

22.4. System state before the use-case starts

The administrator must be logged in.

22.5. System state after the use-case ends

If the use-case succeeds, the account status is updated. Otherwise, the system state does not change.

22.6. Extension points

None.

23. Use-case Search students

23.1. Summary

This use-case allows a teacher or an administrator to search for students in the system.

23.2. Flow of events

Main flow

The use-case starts when the user selects the function to search for students.

1. The system displays a form for the user to enter search criteria (student ID, full name, class, major).
2. The user enters the criteria and requests the search.

3. The system searches and lists the students that match the criteria.
4. If the user selects a student, the use-case Open student detailed record is performed.

Alternative flows

No student matches the criteria: the system displays a no-results message.

23.3. Special requirements

Search results must be paginated when there are many. A teacher only sees students in the course sections assigned to them.

23.4. System state before the use-case starts

The user (Teacher or Administrator) must be logged in.

23.5. System state after the use-case ends

None.

23.6. Extension points

Use-case Open student detailed record: if the user chooses to view a student's detailed record.

24. Use-case Add student record

24.1. Summary

This use-case allows an administrator to add the record of a new student to the system.

24.2. Flow of events

Main flow

The use-case starts when the administrator selects the function to add a student record.

1. The system asks the administrator to enter the student information: student ID, full name, date of birth, gender, address, phone number, email, class and academic status.
2. The administrator enters the information and requests to add the record.

3. The system validates the information and adds the record to the system.
4. The system reports the result of the addition.

Alternative flows

- The student ID already exists: the system displays an error message. The administrator can correct it or cancel.
- Required information is missing or invalid: the system displays an error message and asks for the information again.

24.3. Special requirements

The student ID is unique in the system.

24.4. System state before the use-case starts

The administrator must be logged in.

24.5. System state after the use-case ends

If the use-case succeeds, the new student record is added to the system. Otherwise, the system state does not change.

24.6. Extension points

None.

25. Use-case Edit student information

25.1. Summary

This use-case allows an administrator to edit the information in a student's record.

25.2. Flow of events

Main flow

The use-case starts when the administrator selects the function to edit a student's information.

1. The system displays the student's current information.
2. The administrator edits the information as desired (except the student ID).
3. The administrator requests the update.

4. The system validates the information and updates it in the system.
5. The system reports the result of the update.

Alternative flows

- The administrator cancels: the system does not update the information.
- Invalid information or missing required fields: the system displays an error message. The administrator can correct it or cancel.

25.3. Special requirements

None.

25.4. System state before the use-case starts

The administrator must be logged in.

25.5. System state after the use-case ends

If the use-case succeeds, the student's information is updated. Otherwise, the system state does not change.

25.6. Extension points

None.

26. Use-case Edit academic status

26.1. Summary

This use-case allows an administrator to edit a student's academic status (for example: studying, on leave, withdrawn, graduated).

26.2. Flow of events

Main flow

The use-case starts when the administrator selects the function to edit a student's academic status.

1. The system displays the student's current academic status.
2. The administrator selects the new academic status and enters a note (if any).

3. The administrator requests the update.
4. The system checks that the status change is valid, updates it in the system and records the change history.
5. The system reports the result of the update.

Alternative flows

- The administrator cancels: the system does not update the status.
- The status change is not valid: the system displays an error message. The administrator can choose again or cancel.

26.3. Special requirements

Every academic status change must be logged (editor, time, old status, new status).

26.4. System state before the use-case starts

The administrator must be logged in.

26.5. System state after the use-case ends

If the use-case succeeds, the student's academic status is updated. Otherwise, the system state does not change.

26.6. Extension points

None.

27. Use-case Manage courses

27.1. Summary

This use-case allows an administrator to manage (view, add, edit, delete) the courses in the system.

27.2. Flow of events

Main flow

The use-case starts when the administrator selects the function to manage courses.

1. The system displays the list of courses.
2. If the administrator chooses "Add", the Add course flow is performed.

3. If the administrator chooses "Update" for a course, the Update course flow is performed.
4. If the administrator chooses "Delete" for a course, the Delete course flow is performed.

Add course

1. The system asks the administrator to enter the course information (course code, course name, credits, description).
2. The administrator enters the information and requests to add the course.
3. The system validates the information and adds the course to the system.
4. The system reports the result of the addition.

Update course

1. The system displays the information of the course to be updated.
2. The administrator edits the information as desired and requests the update.
3. The system validates the information and updates it in the system.
4. The system reports the result of the update.

Delete course

1. The system asks the administrator to confirm the deletion of the course.
2. If the administrator confirms, the system checks that the course has no open course section and then deletes the course from the system.
3. The system reports the result of the deletion.

Alternative flows

- The course code already exists or the information is invalid: the system displays an error message. The administrator can correct it or cancel.
- The course already has open course sections or grades: the system reports that it cannot be deleted.

27.3. Special requirements

The course list must be paginated to improve system performance.

27.4. System state before the use-case starts

The administrator must be logged in.

27.5. System state after the use-case ends

If the use-case succeeds, the course information is added, updated or removed from the system. Otherwise, the system state does not change.

27.6. Extension points

None.

28. Use-case Create course sections

28.1. Summary

This use-case allows an administrator to create a new course section for a course so that students can register.

28.2. Flow of events

Main flow

The use-case starts when the administrator selects the function to create a course section.

1. The system asks the administrator to select a course and enter the section information: section code, semester, responsible teacher, schedule, classroom and maximum capacity.
2. The administrator enters the information and requests to create the section.
3. The system validates the information (no conflict with the teacher's or the classroom's schedule) and creates the section.
4. The system reports the result of creating the section.

Alternative flows

- The section code already exists: the system displays an error message. The administrator can correct it or cancel.
- The teacher or the classroom has a schedule conflict: the system displays an error message and asks the administrator to choose again.

28.3. Special requirements

None.

28.4. System state before the use-case starts

The administrator must be logged in and the course must already exist in the system.

28.5. System state after the use-case ends

If the use-case succeeds, the new section is added to the system and appears in the list of open course sections. Otherwise, the system state does not change.

28.6. Extension points

None.

29. Use-case View course section student list

29.1. Summary

This use-case allows an administrator to view the list of students registered for a course section.

29.2. Flow of events

Main flow

The use-case starts when the administrator selects the function to view the student list of a course section.

1. The system asks the administrator to select a course section.
2. The system displays the list of students registered for the section: student ID, full name, class and registration date.
3. The system displays the current enrollment and the maximum capacity of the section.

Alternative flows

No student has registered for the section: the system displays a corresponding message.

29.3. Special requirements

The student list must be paginated when it is large.

29.4. System state before the use-case starts

The administrator must be logged in.

29.5. System state after the use-case ends

None.

29.6. Extension points

None.

Class Diagram

